Question 2		
(a)(i)	T = 24 x 3600 = 86400 s	**A1**
(a)(ii)	Gravitational force provides centripetal force $\frac{GMm}{r^2} = mr\omega^2 = mr(\frac{2\pi}{T})^2$ $r^3 = \frac{GMT^2}{4\pi^2}$ $r = \sqrt[3]{\frac{(6.67\times10^{-11})(5.97\times10^{24})(86400)^2}{4\pi^2}} = 4.22\times10^7$ m	**B1** **A1**
(b)(i)	$\Delta\phi = -\frac{GM}{r_{far}} - (-\frac{GM}{r_{near}}) = GM(\frac{1}{r_{near}} - \frac{1}{r_{far}})$ $\Delta\phi = (6.67\times10^{-11})(5.97\times10^{24})(\frac{1}{6.87\times10^6} - \frac{1}{4.62\times10^7}) = 4.93\times10^7$ J kg^{-1}	**B1** **A1**
(b)(ii)	Loss in KE = Gain in GPE = $m\Delta\phi$ = $1500(4.93\times10^7) = 7.40\times10^{10}$ J Change in KE from closest to furthest = -7.40 x 10^{10} J Note to marker: Deduct 1 mark if a positive answer is given.	**M1** **A1**
(b)(iii)	The satellite moves slowest at its furthest point (due to minimum kinetic energy), meaning it spends the greatest proportion of its orbital time / lingers over the polar region.	**A1**